Problem 1.

Create a countdown timer, where the user is prompted to enter time in seconds and will countdown to zero (set timer delay to 1) using timer.sleep(time_lapse). The program should prompt the user to test the timer if the answer is ‘y’ it will ask the user to enter time in second. If the answer is ‘n’ it will terminate the timer. Your response to y or n should be case insensitive.

Source Code:

```
import time

print("Timer Sample")
answer = 'Y'
while answer == 'Y':
 my_time = int(input("Enter Time is Seconds: "))
for t in range (my_time, 0, -1):
 second = t % 60
 minute = (t // 3600 // 60)
 hour = (t // 3600)
 print(f"{hour:02}:{minute:02}:{second:02}")
time.sleep(1)
 print("Time Is Up!!!\n")
 answer = input("Try Again?[Y/N]: ")
 if answer == 'N':
 print("Bye! Thanks for using the program")
```

Sample output:

```
Timer Sample
Enter Time is Seconds: 10
00:00:10
00:00:09
00:00:08
00:00:07
00:00:06
00:00:05
00:00:04
00:00:03
00:00:02
00:00:01
Time Is Up!!!

Try Again?[Y/N]: Y
Enter Time is Seconds:
```

```
Timer Sample
Enter Time is Seconds: 5
00:00:05
00:00:04
00:00:03
00:00:02
00:00:01
Time Is Up!!!

Try Again?[Y/N]: n
```

Problem 2.

Create an n x n Multiplication table using Nested FOR Loop. The user must enter the number of rows and columns that will be displayed in the Table.

Source code:

```python
rows = int(input("How many rows: "))
cols = int(input("How many cols: "))
print("Multiplication table")

for rows in range(1, rows + 1):
 for cols in range(1, cols + 1):
 print(rows * cols, end="\t")
 print()
```

Output:

```
Multiplication table
1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	12	14	16	18	20
3	6	9	12	15	18	21	24	27	30
4	8	12	16	20	24	28	32	36	40
5	10	15	20	25	30	35	40	45	50
6	12	18	24	30	36	42	48	54	60
7	14	21	28	35	42	49	56	63	70
8	16	24	32	40	48	56	64	72	80
9	18	27	36	45	54	63	72	81	90
10	20	30	40	50	60	70	80	90	100
```

```
How many rows: 3
How many cols: 5
Multiplication table
1   2   3   4   5
2   4   6   8   10
3   6   9   12  15

Process finished with exit code 0
```